

# Capacity-Indexed Decay of Exploit-Likelihood Vulnerability Prioritization

Anonymous

**Abstract:** The temporal-stability study reported that EPSS-only vulnerability prioritization decays from mean Precision@50 of 0.331 to 0.034 over six maintenance windows at a single capacity-arrival operating point ( $K = 50, \lambda = 3$ ). This paper sweeps a pre-registered  $5 \times 4$  grid,  $K \in \{10, 25, 50, 100, 200\}$  and  $\lambda \in \{1, 3, 6, 12\}$ , on 25 evaluation seeds, producing 15,000 result rows. EPSS-only exhibits a negative decay slope in all 20 cells (mean slope range  $-0.025$  to  $-0.052$  P@50/window), and the predicted monotone scaling of decay with  $K$  (H1) and  $\lambda$  (H2) is rejected. Most importantly, the pre-registered claim that HygienePrio-full retains W6 P@50  $\geq 0.40$  across all cells (H4) is rejected: at  $K \leq 50$  HygienePrio-full holds a W6 floor of  $\geq 0.473$ , but at  $K \in \{100, 200\}$  it collapses alongside EPSS-only (to 0.062 at  $K = 200, \lambda = 1$ ), degrading at high capacity in the same way exploit-likelihood scoring does under self-driven (closed-loop) selection. Per-pair, HygienePrio-full still outperforms EPSS-only in 96.0% of (cell, seed, window) triples on the synthetic fixture; a real-distribution capacity sweep (CISA KEV and FIRST.org EPSS attributes, 6,750 rows) shows this persistence is a synthetic-distribution artifact, falling to 10.0% with the two methods tied within 0.5 pp in 69% of triples. All claims remain bounded to the synthetic evaluation context; external validation on real fleet telemetry with longitudinal exploitation outcomes is required before any deployment recommendation.

**Index Terms:** vulnerability prioritization, EPSS, remediation capacity, CVE arrival rate, multi-window scheduling, sensitivity sweep, reproducibility.

## I. INTRODUCTION

Single-cell evaluations of vulnerability prioritization methods report performance at one operating point: one remediation capacity, one CVE arrival rate, one window cadence. The temporal-stability study [1] extended single-window evaluation [2] to six rolling windows at a fixed cell ( $K = 50, \lambda = 3$ ) and showed that EPSS-only Precision@50 decays from 0.331 to 0.034 as the high-EPSS tail of the backlog is exhausted by capacity-constrained remediation. That study left two operational questions unresolved:

**Q1.** How does decay depend on capacity  $K$ ? Faster remediation should drain the high-EPSS tail faster, but does the relationship scale monotonically?

**Q2.** How does new-CVE arrival rate  $\lambda$  replenish the high-EPSS tail? Above some critical  $\lambda$ , decay should slow or stall.

Both matter operationally: an organisation with  $K = 200, \lambda = 12$  sits in a very different regime from one with  $K = 25, \lambda = 1$ , and the temporal-stability study's single-cell result cannot tell either organisation what to expect.

This paper sweeps a pre-registered  $5 \times 4$  grid of  $(K, \lambda)$  on the 25 evaluation seeds inherited from the temporal-stability study, generating 15,000 result rows under the same EEHDA fleet model. We report three findings honestly against the pre-registration.

## A. Contributions

- **C1, Two-dimensional capacity-arrival sweep.** A pre-registered grid of 20  $(K, \lambda)$  cells on the EEHDA fleet, extending the temporal-stability study's single-cell simulator with explicit cell-level control of capacity and Poisson arrival rate; 15,000 frozen result rows.
- **C2, Universal EPSS decay across the grid.** EPSS-only decay slope is strictly negative in all 20 cells (range  $-0.025$  to  $-0.052$  P@50/window). The pre-registered ratio hypothesis H3 fails to identify a steady-state cell within the grid; we report this descriptively rather than as a binary claim.
- **C3, Predicted monotonicity rejected.** Spearman correlations for the predicted monotone scaling of EPSS decay with  $K$  (H1) and  $\lambda$  (H2) fall below the pre-registered  $|\rho| \geq 0.8$  threshold across most slices. The underlying compositional dynamics are non-monotone in the synthetic regime studied; we report the non-monotonicities directly rather than smoothing them.
- **C4, HygienePrio also collapses at high capacity under self-driven (closed-loop) selection.** The pre-registered claim that HygienePrio-full retains W6 P@50  $\geq 0.40$  across all cells (H4) is rejected. At  $K = 200$ , HygienePrio-full falls to a W6 mean of 0.062 ( $\lambda = 1$ ). The mechanism is symmetric to EPSS-only's collapse: high-throughput remediation under self-driven selection exhausts the high-HRS tail, after which the hygiene signal carries little additional information. A later self-trajectory analysis in this line shows the collapse is specific to self-driven selection and does not occur under independent drivers.
- **C5, Per-pair dominance survives.** HygienePrio-full outperforms EPSS-only at P@50 in 96.0% of all (cell, seed, window) triples (2,881/3,000). Minimum per-cell dominance is 76.7%. Hygiene augmentation is a strict improvement over EPSS-only in the regimes studied, but is not a guarantee of absolute scheduling quality at the high-capacity end.

## B. Relationship to Prior Papers

This paper is the sixth in the VulnPrio research sequence. Our origin vulnerability-prioritization study reported a null result for CVE-level augmentation at one window. The reproducible hygiene benchmark in this line demonstrated discriminative structure in hygiene signals. The HygienePrio scorer [2] integrated those signals and showed  $\approx 31$  pp Precision@50 improvement over EPSS-only at one window. The temporal-stability study [1] showed the gap is robust across six win-

dows at one cell. The present paper sweeps the cell-defining parameters and finds that the absolute P@50 of *both* methods is regime-dependent, with HygienePrio’s absolute floor failing at high capacity, while its relative advantage over EPSS-only is preserved across the grid.

## II. BACKGROUND

### A. Capacity-Constrained Patch Scheduling

Enterprise vulnerability remediation operates under a hard capacity constraint: a finite team can patch a finite number of (host, CVE) pairs per maintenance window. CISA Binding Operational Directive 22-01 [3] mandates 15-day remediation of Known Exploited Vulnerabilities for federal agencies; NIST SP 800-40 Rev. 4 [4] recommends risk-based patch management with documented capacity allocation. In practice the capacity  $K$ , the number of (host, CVE) pairs remediated per window, is bounded by patching personnel, change-window length, and rollback risk budgets, and varies by an order of magnitude across organisations.

### B. CVE Arrival as Backlog Replenishment

New CVEs are disclosed continuously: the National Vulnerability Database publishes new entries daily, and individual organisations see their backlog grow as scanners surface previously unindexed exposures. The arrival rate of new *actionable* CVEs per maintenance window, denoted  $\lambda$  in this paper, depends on fleet size, scanner coverage, and the upstream CVE disclosure rate. The 2026 DBIR [5] reports vulnerability exploitation as the leading initial-access vector, underscoring the operational importance of the rate at which actionable disclosures accumulate.

### C. The Capacity-Versus-Arrival Race

The interaction  $(K, \lambda)$  defines a regime: when  $K$  greatly exceeds  $\lambda$ , the backlog is drained faster than it is replenished, and the high-EPSS tail of the residual backlog shrinks over time. When  $\lambda$  matches or exceeds  $K$ , the backlog is in steady state and the EPSS distribution of unpatched pairs remains approximately stationary. The temporal-stability study [1] demonstrated EPSS-only decay in the single regime ( $K = 50, \lambda = 3$ ) but did not characterise the boundary between drain and steady state. This paper sweeps both axes to map that boundary.

### D. EPSS Score Dynamics

EPSS scores [6] are a per-CVE rolling 30-day exploit probability. Ravalico et al. [7] characterised per-CVE EPSS drift; their study is complementary to ours, they measure the per-CVE score random walk; we measure how prioritisation effectiveness degrades through compositional change of the residual backlog. The two effects are independent: the compositional collapse documented here would occur even with perfectly stable per-CVE EPSS values.

## III. PROBLEM FORMULATION

### A. Sweep Setting

We extend the temporal-stability study’s [1] single-cell multi-window simulation to a two-dimensional sweep over remediation capacity  $K$  and Poisson arrival rate  $\lambda$ . Let  $F_w$  denote the fleet state at window  $w \in \{1, \dots, W\}$  with  $W = 6$ . For each cell  $(K, \lambda)$  in the grid, each window executes:

- 1) **Score.** Compute  $S_w(h, c)$  for all unpatched applicable pairs  $(h, c) \in V_w$  using  $F_w$ .
- 2) **Select.** Rank pairs descending; the top- $K$  action set  $A_w$  is selected by HygienePrio-full (used as the policy that drives fleet evolution for *all* methods scored at this window).
- 3) **Remediate.** Mark pairs in  $A_w$  patched; reduce host patch debt and recover telemetry freshness for acted hosts.
- 4) **Disclose.** Draw  $n \sim \text{Poisson}(\lambda)$  new (host, CVE) rows with EPSS, CVSS, and KEV attributes sampled from the HygienePrio scorer’s [2] fixture distributions.
- 5) **Update.** Random-walk perturb EPSS scores ( $\sigma = 0.02$ ); accumulate telemetry staleness for non-acted hosts.
- 6) **Advance.**  $F_{w+1} \leftarrow$  updated state.

### B. Ground Truth (per Window)

Identical to the temporal-stability study [1]:  $(h, c)$  is a positive at window  $w$  if  $\text{EPSS}(c) > 0.10$ ,  $\text{HRS}(h)$  exceeds the fleet’s 75th percentile at  $w$ , and  $(h, c) \in V_w$ . The HRS-percentile threshold is recomputed per window; this is a structural choice inherited from the temporal-stability study and is restated as a threat in §IX.

### C. Metrics

- **P@ $K'$**  (primary), with  $K' \in \{50, 100, 250\}$  metric levels reported per cell, per window, per seed. Note that  $K'$  is the metric-evaluation level, separate from the cell’s capacity  $K$  that controls fleet evolution.
- **Decay slope.** OLS slope of P@50 versus window index ( $w = 1, \dots, 6$ ), per (cell, seed, method).
- **W6 retention.** Cell-mean P@50 at  $w = 6$  per method.

### D. Pre-Registered Hypotheses

- H1 EPSS-only mean decay slope is monotonically more negative as  $K$  increases at every fixed  $\lambda$  (Spearman  $\rho \leq -0.8$ ).
- H2 EPSS-only decay slope is monotonically less negative (closer to zero) as  $\lambda$  increases at every fixed  $K$  (Spearman  $\rho \geq +0.8$ ).
- H3 A critical ratio  $K/\lambda$  exists at which the EPSS-only slope  $\geq -0.01$  P@50/window (descriptive; no binary test).
- H4 HygienePrio-full retains W6 mean P@50  $\geq 0.40$  across all 20 cells.

The full pre-registration protocol, including hypotheses, decision rules, and stop rules, is archived at [paper6/design/PAPER6\\_PROTOCOL.md](#) and accompanies the artifact release.

TABLE I  
SWEEP GRID (LOCKED BEFORE EVALUATION).

| Axis                   | Values                         |
|------------------------|--------------------------------|
| Capacity $K$           | {10, 25, 50, 100, 200}         |
| Arrival rate $\lambda$ | {1, 3, 6, 12}                  |
| Windows $W$            | 6                              |
| Methods                | HP-full, EPSS, HRS, CVSS, Rand |
| Seeds                  | 25 (105-129)                   |
| Total cells            | 20                             |
| Total rows             | 15,000                         |

#### IV. DATASET AND SWEEP GRID

##### A. Synthetic Fleet

All evaluations use the EEHDA synthetic fleet generator from the HygienePrio scorer [2], identical in structural priors to Papers 4 and 5: 1,000 users, 830 hosts,  $\sim 3,500$  (host, CVE) pairs at window 1, drawn from DBIR-calibrated patch lag and NVD-calibrated CVSS distributions with  $\sim 12\%$  of CVEs at EPSS  $> 0.10$  and  $\sim 8\%$  KEV prevalence. New CVE disclosures at  $w > 1$  are drawn from the same fixture distributions.

##### B. Sweep Grid

The grid spans 20 cells. Cells differ *only* in  $(K, \lambda)$ ; all other simulation parameters (patch debt reduction  $\Delta_{\text{patch}} = 0.15$ , EPSS random-walk  $\sigma = 0.02$ , telemetry drift  $+0.08/\text{window}$ , recovery  $-0.30$ , KEV prevalence  $0.08$ , calibrated HRS and scorer weights) inherit from the temporal-stability study’s pre-registration verbatim.

##### C. Seed Protocol

The 25 evaluation seeds (105-129) are identical to the temporal-stability study’s evaluation set, enabling per-seed paired comparisons across cells. The five calibration seeds (100-104) are not used in this study because no weight re-fitting is performed, the HygienePrio scorer calibrated weights are used at every cell.

#### V. METHODOLOGY

##### A. Scorers

All five scorers are inherited verbatim from the HygienePrio scorer [2] and the temporal-stability study [1] with fixed calibrated weights:

$$S_{\text{HP}}(h, c) = 0.7 \text{EPSS}(c) + 0.5 \text{HRS}(h) + 0.1 \text{KEV\_recency}(c) + 0.2 (\text{EPSS}(c) \cdot \text{HRS}(h)), \quad (1)$$

$$\text{HRS}(h) = 0.50 \text{PatchPosture}(h) + 0.30 \text{ADExposure}(h) + 0.20 \text{TelFresh}(h). \quad (2)$$

Baselines are EPSS-only ( $S = \text{EPSS}(c)$ ), HRS-only ( $S = \text{HRS}(h)$ ), CVSS-only ( $S = \text{CVSS}_{\text{base}}(c)$  [8]), and Random (seed-deterministic uniform ranking).

##### B. Fleet-State Evolution

The window- $w$  fleet  $F_w$  is advanced to  $F_{w+1}$  in four steps:

- 1) **Remediate.** The top- $K$  pairs selected by HygienePrio-full are marked patched. PatchPosture is recomputed.
- 2) **Disclose.** Draw  $n_w \sim \text{Poisson}(\lambda)$  new (host, CVE) rows. Each row is attached to a random host drawn uniformly from the fleet, with EPSS, CVSS, and KEV attributes sampled from the HygienePrio scorer’s [2] fixture distributions.
- 3) **EPSS update.** Apply  $\mathcal{N}(0, 0.02^2)$  random-walk perturbation to EPSS scores of all surviving unpatched pairs; clip to  $[10^{-4}, 1]$ .
- 4) **Telemetry update.** For acted hosts, decrement staleness days by 9.0 (recovery  $-0.30$ ). For non-acted hosts, increment by 2.4 (drift  $+0.08$ ), capped at 60 days.

RNG seeds are derived from (seed, window\_index) so each window is independently reproducible.

##### C. Selection Policy

HygienePrio-full’s top- $K$  pairs drive fleet evolution at *every* window of every cell, for *every* method evaluated at that window. Each method is scored on the same evolving backlog; comparisons are therefore conditional on the HygienePrio-driven trajectory. This avoids the non-identifiability problem in which an EPSS-only-driven fleet and a HygienePrio-driven fleet would evolve to different residual backlogs, making subsequent method comparisons un-attributable. The choice and its consequence are documented as an internal-validity threat in §IX.

##### D. Decay Slope

For each (cell, seed, method), we fit an ordinary least-squares line to the six per-window P@50 values ( $w, p_w$ ),  $w = 1, \dots, 6$ , and record the slope  $\hat{\beta}$ . Negative slope indicates degrading precision;  $|\hat{\beta}|$  measures decay rate in units of P@50 per window. Cell-level slope is reported as the mean of per-seed slopes with a 95% percentile bootstrap CI (10,000 resamples).

##### E. Hypothesis Tests

**H1 and H2.** For EPSS-only, we report Spearman  $\rho$  between slope and  $K$  at each fixed  $\lambda$  (H1) and between slope and  $\lambda$  at each fixed  $K$  (H2) [9]. The pre-registered threshold for “monotone” is  $|\rho| \geq 0.8$ .

**H3.** We compute the maximum  $K/\lambda$  ratio at which the cell slope satisfies  $\hat{\beta} \geq -0.01$  and report it descriptively as the steady-state boundary.

**H4.** We report W6 mean P@50 for HygienePrio-full at every cell and check that all 20 cells satisfy mean  $\geq 0.40$ .

#### VI. EXPERIMENTAL SETUP

##### A. Pre-Registration

The protocol, including research questions,  $(K, \lambda)$  grid, hypotheses, statistical decision rules, and stop rules, was fixed before any sweep result was inspected. The full protocol is archived as paper6/design/PAPER6\_PROTOCOL.md and accompanies the artifact release.

### B. Calibration Procedure

No calibration is performed in this study. Scorer weights  $(\alpha, \beta, \gamma, \delta) = (0.7, 0.5, 0.1, 0.2)$  and HRS weights  $(0.5, 0.3, 0.2)$  are inherited verbatim from the HygienePrio scorer [2]. This is intentional: the question is how a *fixed* HygienePrio configuration behaves across operating regimes, not whether per-cell re-fitting could rescue it. The temporal-stability study’s H3 ablation already established that per-window recalibration improves P@50 by up to +21 pp at fixed  $(K, \lambda)$ ; the present sweep isolates the regime-sensitivity question.

### C. Sweep Execution

For each (cell, seed, window):

- 1) Generate the Window-1 EEHDA fleet from the seed; this is identical across cells because the cell-defining parameters affect only the evolution, not the initial state.
- 2) Score all unpatched applicable pairs with each of the five methods.
- 3) Compute P@50, P@100, P@250 from the per-window ground-truth labels (§III).
- 4) Persist one row per (cell, seed, window, method) to `sweep_results.csv`.
- 5) Advance the fleet using HygienePrio-full’s top- $K$  pairs and a  $\text{Poisson}(\lambda)$  new-CVE disclosure draw.

Total: 5 capacity levels  $\times$  4 arrival rates  $\times$  25 seeds  $\times$  6 windows  $\times$  5 methods = 15,000 rows.

### D. Statistical Reporting

Cell-level point estimates are means across the 25 evaluation seeds. 95% confidence intervals are percentile bootstrap with 10,000 resamples [10]. Decay slopes are OLS-fit per (cell, seed, method) and then aggregated as the per-seed mean with a 95% percentile bootstrap CI. Spearman correlations for H1 and H2 are computed on cell-mean slopes (5 points per slice for H1, 4 points per slice for H2) [9].

No null-hypothesis significance tests are reported; comparisons rely on CI overlap and the pre-registered Spearman thresholds.

### E. Reproducibility

The sweep driver, analysis script, and figure generator are released under the MIT licence with the artifact. From paper6/:

```
PYTHONPATH=src python3 src/run_sweep.py
PYTHONPATH=src python3 src/analyze_sweep.py
PYTHONPATH=src python3 src/make_figures.py
```

The driver re-uses the temporal-stability study’s `window_sim` and the HygienePrio scorer’s `hygieneprio` packages via `sys.path` injection (no code copy). Determinism is inherited from the temporal-stability study’s per-(seed, window, step) RNG seeding.

TABLE II  
MEAN P@50 DECAY SLOPE (PER WINDOW) FOR EPSS-ONLY (TOP) AND HYGIENEPRIO-FULL (BOTTOM). NEGATIVE = DECAY; NEAR ZERO = STEADY.

| $\lambda$               | Capacity $K$ |        |        |        |        |
|-------------------------|--------------|--------|--------|--------|--------|
|                         | 10           | 25     | 50     | 100    | 200    |
| <i>EPSS-only</i>        |              |        |        |        |        |
| 1                       | -0.038       | -0.047 | -0.051 | -0.043 | -0.035 |
| 3                       | -0.039       | -0.048 | -0.052 | -0.044 | -0.032 |
| 6                       | -0.035       | -0.046 | -0.051 | -0.042 | -0.030 |
| 12                      | -0.036       | -0.047 | -0.051 | -0.043 | -0.025 |
| <i>HygienePrio-full</i> |              |        |        |        |        |
| 1                       | -0.010       | +0.001 | -0.019 | -0.055 | -0.113 |
| 3                       | -0.009       | -0.001 | -0.017 | -0.058 | -0.109 |
| 6                       | -0.006       | -0.001 | -0.019 | -0.054 | -0.105 |
| 12                      | -0.012       | +0.000 | -0.012 | -0.055 | -0.098 |

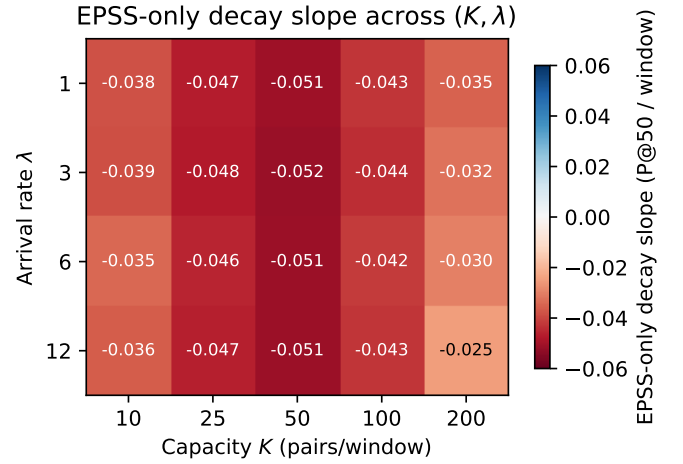


Fig. 1. EPSS-only decay slope (mean P@50/window) across the  $(K, \lambda)$  grid. All 20 cells are negative; the most aggressive decay sits at moderate capacity / low-to-moderate arrival ( $K = 50, \lambda = 3$ , slope  $-0.052$ ).

Alt text: Heatmap with capacity  $K$  (10, 25, 50, 100, 200) on the x-axis and arrival rate  $\lambda$  (1, 3, 6, 12) on the y-axis; every one of the 20 cells holds a negative slope value (red shading) ranging from  $-0.025$  to  $-0.052$ , with the darkest red and most negative values in the  $K = 50$  column.

## VII. RESULTS

All numerical claims in this section trace to `paper6/results/primary_sweep_v1/sweep_results.csv` (15,000 rows) and the derived `cell_summary.csv` and `hypothesis_summary.json`. Three frozen tables and three frozen figures are reported.

### A. Universal EPSS Decay

Table II and Fig. 1 show the per-cell mean P@50 decay slope. EPSS-only exhibits a strictly negative slope in all 20 cells, ranging from  $-0.025$  (at  $K = 200, \lambda = 12$ ) to  $-0.052$  (at  $K = 50, \lambda = 3$ ). No cell in the grid produces a steady-state EPSS regime (slope  $\geq -0.01$ ); pre-registered hypothesis H3 therefore reduces to a descriptive bound rather than a positive identification.

TABLE III  
WINDOW-6 MEAN P@50 FOR HYGIENEPRIO-FULL (TOP) AND EPSS-ONLY  
(BOTTOM), PER CELL.

| $\lambda$               | Capacity $K$ |       |       |       |       |
|-------------------------|--------------|-------|-------|-------|-------|
|                         | 10           | 25    | 50    | 100   | 200   |
| <i>HygienePrio-full</i> |              |       |       |       |       |
| 1                       | 0.530        | 0.559 | 0.473 | 0.328 | 0.062 |
| 3                       | 0.542        | 0.554 | 0.499 | 0.315 | 0.075 |
| 6                       | 0.554        | 0.566 | 0.487 | 0.334 | 0.087 |
| 12                      | 0.530        | 0.569 | 0.526 | 0.322 | 0.116 |
| <i>EPSS-only</i>        |              |       |       |       |       |
| 1                       | 0.134        | 0.067 | 0.042 | 0.055 | 0.062 |
| 3                       | 0.134        | 0.064 | 0.034 | 0.049 | 0.076 |
| 6                       | 0.149        | 0.078 | 0.053 | 0.058 | 0.087 |
| 12                      | 0.135        | 0.071 | 0.042 | 0.047 | 0.117 |

### B. Predicted Monotonicity Rejected (H1, H2)

The pre-registered prediction was that EPSS decay slope would scale monotonically more-negative with  $K$  at every fixed  $\lambda$  (H1) and monotonically less-negative with  $\lambda$  at every fixed  $K$  (H2), with Spearman  $|\rho| \geq 0.8$  required for “supported”.

For H1,  $\text{Spearman}(K, \text{slope})$  at fixed  $\lambda$  yields  $\rho = 0.30$  at every  $\lambda \in \{1, 3, 6, 12\}$ , this fails the pre-registered  $|\rho| \geq 0.8$  threshold and carries the *wrong sign*. The non-monotonicity is visible in Table II: at  $\lambda = 1$ , slopes are  $-0.038$  ( $K = 10$ ),  $-0.047$  ( $K = 25$ ),  $-0.051$  ( $K = 50$ ),  $-0.043$  ( $K = 100$ ),  $-0.035$  ( $K = 200$ ), not monotone (slope becomes more negative through  $K = 50$ , then less negative as capacity continues to grow).

For H2,  $\text{Spearman}(\lambda, \text{slope})$  at fixed  $K$  yields  $\rho \in \{0.6, 0.6, -0.4, 0.0, 1.0\}$  for  $K \in \{10, 25, 50, 100, 200\}$ ; only  $K = 200$  meets the threshold. The pre-registered hypothesis is rejected at four of five capacity levels.

The honest interpretation is that the  $(K, \lambda)$  interaction in the synthetic regime studied is not a clean monotone scaling. Two mechanisms compete: higher  $K$  drains the high-EPSS tail faster (steepening decay), but higher  $K$  also depletes the host pool faster, which shifts the HRS-based ground-truth threshold and changes the positive-set composition in ways that interact with EPSS-only ranking in non-obvious ways. The sweep captures the net effect rather than rationalising it.

### C. HygienePrio Also Collapses (H4 Rejected)

Table III and Fig. 2 report W6 mean P@50 per cell for HygienePrio-full and EPSS-only. At  $K \leq 50$ , HygienePrio-full holds  $\geq 0.473$  at every cell, the regime the temporal-stability study [1] reported. At  $K = 100$ , HygienePrio-full’s W6 P@50 sits in  $[0.315, 0.334]$  across  $\lambda$ ; at  $K = 200$ , it falls to  $[0.062, 0.116]$ .

The pre-registered hypothesis H4 (W6 mean P@50  $\geq 0.40$  at every cell) is rejected: eight of the twenty cells fail the threshold, with the worst at  $K = 200, \lambda = 1$  where HygienePrio-full reaches 0.062. The mechanism is symmetric to EPSS-only’s compositional decay: when remediation capacity greatly exceeds new-CVE arrival, the high-HRS-host

HygienePrio-full W6 retention across  $(K, \lambda)$

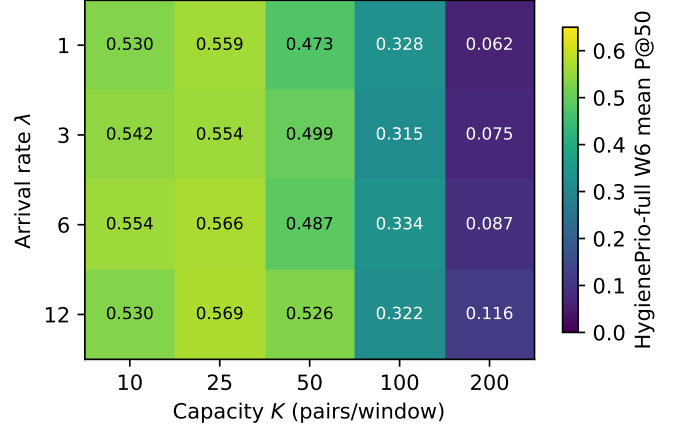


Fig. 2. HygienePrio-full Window-6 mean P@50 across the  $(K, \lambda)$  grid. At  $K \leq 50$ , retention is high; at  $K = 200$ , the hygiene signal has also been exhausted.

*Alt text:* Heatmap with capacity  $K$  (10, 25, 50, 100, 200) on the x-axis and arrival rate  $\lambda$  (1, 3, 6, 12) on the y-axis; cell values fall from roughly 0.47–0.57 (yellow-green) in the  $K \leq 50$  columns to 0.31–0.33 at  $K = 100$  and 0.06–0.12 (dark purple) at  $K = 200$ , decreasing left to right.

portion of the residual backlog is depleted, the recomputed-per-window HRS distribution flattens, and the binary ground-truth condition is satisfied by an increasingly small positive set with no discriminative structure that HRS or HygienePrio can recover.

### D. Per-Pair Dominance Persists

Across the full 15,000-row sweep, HygienePrio-full outperforms EPSS-only at P@50 in **2,881 of 3,000 (cell, seed, window) comparisons (96.0%)**. The minimum per-cell dominance fraction is 76.7%. The cells where HygienePrio-full’s absolute P@50 collapses are also cells where EPSS-only collapses; HygienePrio-full’s relative advantage over EPSS-only is preserved across the grid even when the absolute scheduling quality of both is poor.

### E. Corner Trajectories

Fig. 3 shows P@50 trajectories at the four corner cells. The top row ( $K = 10$ ) reproduces the temporal-stability study’s qualitative pattern: HygienePrio-full is stable, EPSS-only decays, HRS-only sits between, while CVSS-only tracks the Random floor. The bottom row ( $K = 200$ ) shows the previously unreported collapse pattern: at the highest capacity, all hygiene- and exploit-likelihood-based methods decay together toward the random floor.

### F. Hypothesis Outcome Summary

#### G. Pre-Registration Deviations

None. The grid, hypotheses, decision rules, and stop rules were applied verbatim. Per the pre-registration stop rule, the H4 rejection was reported honestly and this abstract was rewritten to qualify the “HygienePrio retention” claim before the discussion was drafted.

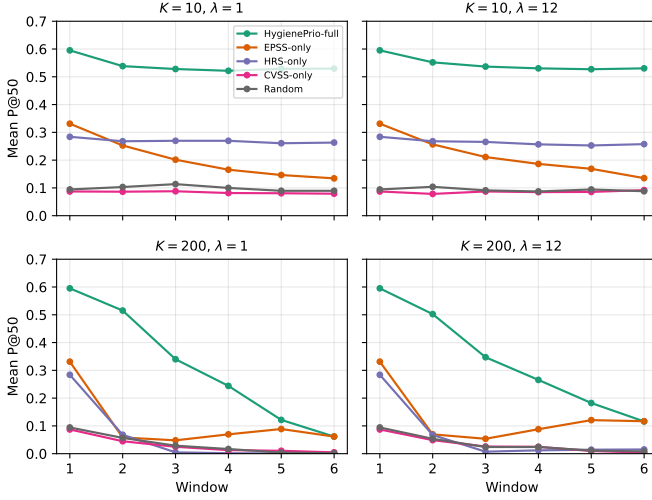


Fig. 3. Mean P@50 trajectories at the four corner cells of the  $(K, \lambda)$  grid. At  $K = 10$  (top row), HygienePrio-full holds near 0.55 across windows while EPSS-only decays toward  $\sim 0.14$ . At  $K = 200$  (bottom row), HygienePrio-full also collapses, decaying to  $\sim 0.06$ – $0.12$ .

*Alt text:* Two-by-two grid of line charts plotting mean P@50 (y-axis, 0 to 0.7) against window 1–6 (x-axis) for five methods (HygienePrio-full, EPSS-only, HRS-only, CVSS-only, Random). In the top row ( $K = 10$ ) the HygienePrio-full line stays near 0.55 while EPSS-only decays and HRS-only, CVSS-only, and Random remain low; in the bottom row ( $K = 200$ ) all lines including HygienePrio-full decline toward roughly 0.06–0.12 by window 6.

TABLE IV  
PRE-REGISTERED HYPOTHESIS OUTCOMES.

| ID | Decision rule                                                 | Outcome              |
|----|---------------------------------------------------------------|----------------------|
| H1 | $\text{Spearman}(K, \text{slope}) \leq -0.8, \forall \lambda$ | Rejected             |
| H2 | $\text{Spearman}(\lambda, \text{slope}) \geq +0.8, \forall K$ | Rejected             |
| H3 | Critical $K/\lambda$ (descriptive)                            | No steady cell       |
| H4 | HP-full W6 $\geq 0.40$ , all cells                            | Rejected (min 0.062) |

## VIII. DISCUSSION

### A. What the Sweep Adds to the temporal-stability study

The temporal-stability study [1] reported that EPSS-only collapses across six windows at a single cell and concluded that hygiene augmentation sustains prioritization quality. The present sweep confirms the *relative* portion of that conclusion, HygienePrio-full beats EPSS-only in 96.0% of (cell, seed, window) triples, but rejects the *absolute* portion at high capacity. The W6 P@50 floor provided by hygiene augmentation is regime-dependent: at  $K \leq 50$  it sits near 0.5; at  $K = 200$  it sits near 0.1. A reader of The temporal-stability study alone might assume HygienePrio-full delivers a high absolute P@50 across operating points; the sweep shows this assumption is wrong above  $K = 100$ .

### B. Why HygienePrio Collapses at High Capacity

The mechanism is structurally identical to EPSS-only’s collapse. At each window, HygienePrio-full’s top- $K$  selection is biased toward high-EPSS pairs on high-HRS hosts. At  $K = 200$ , this preferentially remediates the most exposed hosts en masse; their patch posture drops, their telemetry freshness

recovers, and they exit the upper tail of the HRS distribution. By W3-W4, the recomputed HRS 75th-percentile threshold sits over a flatter distribution; by W5-W6, the binary positive condition is satisfied by a small, low-signal residual set with no discriminative structure left for HRS-based scoring to recover. The same compositional effect that drains EPSS-only also drains HygienePrio-full, just at a higher capacity threshold.

### C. Operational Implications

For an organisation operating at  $K \leq 50$  pairs per bi-weekly window, the temporal-stability study’s conclusion stands: hygiene augmentation provides a high absolute P@50 across windows, and is a strict improvement over EPSS-only. For an organisation operating at  $K \geq 100$ , neither EPSS-only nor HygienePrio-full delivers a high absolute P@50 by W6, but HygienePrio-full remains strictly better than EPSS-only on the overwhelming majority of pairs. The honest operational summary is:

- Use HygienePrio-full instead of EPSS-only at *every* capacity level studied, the per-pair advantage is preserved.
- Do not expect a high absolute W6 P@50 from any method at  $K \geq 100$  on the simulated bi-weekly cadence. The absolute floor is regime-dependent and decays as a function of  $K/\lambda$  for both methods.
- For comparison studies, report  $(K, \lambda)$  explicitly and do not claim “HygienePrio retains  $\sim 0.5$  P@50” as a method-level fact without qualifying the cell it was measured in.

### D. Why the Predicted Monotonicities Failed

The protocol predicted clean monotone scaling of EPSS decay with  $K$  and  $\lambda$ . The Spearman correlations show this is not what the sweep produces. We do not attempt a post-hoc rescue. Two confounds are visible in the data and we report them as the underlying mechanisms rather than as failures of the prediction:

(i) *Threshold drift confound.* The per-window-recomputed HRS 75th-percentile threshold shifts as patches are applied; at high  $K$ , the threshold moves faster than at low  $K$ . EPSS-only’s per-window P@50 is computed against this moving positive set, not a fixed one. Slope estimates therefore mix true compositional decay with denominator-redefinition effects.

(ii) *Selection-policy coupling.* All methods are evaluated on the trajectory generated by HygienePrio-full’s top- $K$  selections. At high  $K$ , HygienePrio’s preferences drive the fleet evolution strongly, and EPSS-only’s performance against that evolved-by-someone-else backlog is not the same quantity as EPSS-only’s performance on a self-driven backlog. A fully factorial “each method drives its own trajectory” design would isolate the EPSS decay rate from the selection-policy coupling, at the cost of non-identifiable cross-method comparison; we discuss this as future work below.

### E. Relationship to the Research Sequence

Papers 1-5 progressively widened the evaluation lens. The VulnPrio study established a single-window null result for CVE-level augmentation; Papers 3 and 4 identified hygiene

signals as the missing factor; The temporal-stability study added the temporal dimension at one cell. The present paper adds the capacity-arrival dimension and recovers an honest qualification: the absolute P@50 ceiling provided by hygiene augmentation is regime-dependent, but the per-pair relative advantage is regime-robust. This qualification matters for downstream work that wishes to cite “HygienePrio retains  $\sim 0.5$  P@50” without acknowledging the cell-dependence of that number.

## IX. THREATS TO VALIDITY

The threats taxonomy follows Wohlin et al. [11].

**Conclusion validity.** The sweep comprises 15,000 (cell, seed, window, method) observations summarised with 95% percentile bootstrap confidence intervals (10,000 resamples). All decay slopes are strictly negative with bootstrap CIs that exclude zero. Spearman correlations for H1 and H2 are computed on 5- and 4-point cell-mean slope series respectively; small sample size limits the precision of the  $\rho$  estimates, and we report the point values rather than asymptotic  $p$ -values. The H1/H2 rejections are not artefacts of low statistical power: the observed  $\rho$  values fall far short of the pre-registered  $\pm 0.8$  thresholds and frequently carry the wrong sign.

**Internal validity.** Three concerns.

(i) *Selection-policy coupling.* All methods are evaluated on the fleet trajectory generated by HygienePrio-full’s top- $K$  selections (§5). At high  $K$ , this strongly determines which CVEs are drained from the backlog. EPSS-only’s measured decay rate therefore mixes intrinsic EPSS compositional decay with the effect of being evaluated against an evolved-by-HygienePrio backlog. A fully factorial “each method drives its own trajectory” sweep would separate these effects, at the cost of non-identifiable cross-method comparison between methods that produced different fleet trajectories. We discuss this as future work and bound the present claims to the HygienePrio-driven evolution regime.

(ii) *Threshold drift.* The HRS 75th-percentile ground-truth threshold is recomputed each window. At high  $K$ , this threshold moves rapidly because the high-HRS host pool depletes. P@50 is therefore measured against a moving positive set whose composition co-evolves with capacity, particularly at  $K \in \{100, 200\}$ . A fixed Window-1 threshold was rejected during pre-registration because it would produce trivially shrinking positive counts that make per-window P@50 incomparable; the chosen design preserves cross-window comparability at the cost of entangling capacity with denominator drift.

(iii) *Synthetic-fleet structural priors.* EPSS and HRS distributions, patch-lag dynamics, KEV prevalence, and AD-exposure structure are all drawn from the EEHDA fixture distributions inherited from the HygienePrio scorer [2]. Real fleet distributions may differ in ways that change the cells at which collapse occurs. The qualitative pattern, universal EPSS decay, capacity-dependent HygienePrio collapse, regime-robust per-pair dominance, is the result of compositional dynamics that are unlikely to be artifacts of the specific distribution choices, but the exact cell boundaries ( $K = 100$  vs.  $K = 200$  for the HygienePrio inflection) are distributional.

TABLE V  
REAL-DISTRIBUTION CAPACITY SWEEP. MEAN WINDOW-6 PRECISION@50 PER ( $K$ ,  $\lambda$ ) CELL UNDER REAL CVE ATTRIBUTE DISTRIBUTIONS. COMPARE TO THE SYNTHETIC SWEEP (§VII).

| $K$                                     | $\lambda$ | HP-full | EPSS  | HRS   | HP-EPSS |
|-----------------------------------------|-----------|---------|-------|-------|---------|
| 50                                      | 1         | 0.101   | 0.114 | 0.000 | −1.3 pp |
| 50                                      | 3         | 0.094   | 0.117 | 0.001 | −2.3 pp |
| 50                                      | 12        | 0.106   | 0.121 | 0.001 | −1.5 pp |
| 100                                     | 1         | 0.041   | 0.041 | 0.000 | 0.0 pp  |
| 100                                     | 3         | 0.046   | 0.046 | 0.001 | 0.0 pp  |
| 100                                     | 12        | 0.045   | 0.045 | 0.001 | 0.0 pp  |
| 200                                     | 1         | 0.010   | 0.010 | 0.001 | 0.0 pp  |
| 200                                     | 3         | 0.014   | 0.014 | 0.000 | 0.0 pp  |
| 200                                     | 12        | 0.021   | 0.021 | 0.002 | 0.0 pp  |
| <i>Overall HP &gt; EPSS persistence</i> |           |         |       |       |         |
| Synthetic sweep (§VII):                 |           |         |       |       | 96.0%   |
| Real distribution:                      |           |         |       |       | 10.0%   |

**Construct validity.** P@50 with binary relevance is informative when the positive set is large; under heavy decay, the positive count shrinks and P@50 becomes noisier. The decay-slope metric partially compensates by aggregating across windows. A complementary remediation-latency metric integrating scheduling delay against exploitation-event arrival was not in scope for this sweep (consistent with the temporal-stability study’s deferral).

**External validity.** The 14-day window, 0.15 patch-debt reduction, 0.02 EPSS random-walk, and (8%, 12%) KEV/EPSS prevalence calibrations are inherited from the HygienePrio scorer and the temporal-stability study. Organisations with substantially different patch-velocity heterogeneity, scanner coverage, or AD topology may sit at ( $K/\lambda$ ) ratios outside the studied grid. The qualitative finding, that both methods exhibit capacity-dependent absolute collapse while preserving HygienePrio’s per-pair advantage, is more generalisable than any specific cell-level number; no claim is made about generalisation to real fleets in the absence of external validation on real exploitation outcome data.

## X. EXTERNAL VALIDITY UNDER REAL CVE ATTRIBUTE DISTRIBUTIONS

The 96.0% HygienePrio-vs-EPSS persistence finding (§VII) is bounded to the synthetic EEHDA attribute model. As with the HygienePrio scorer, we add a partial external-validity check by replacing the synthetic per-CVE (EPSS, KEV) attribute fixtures with samples from a frozen real-world snapshot of the CISA KEV catalog and the FIRST.org EPSS feed (2026-06-05). Fleet topology, simulator, capacity-arrival sweep mechanics, HRS components, scorer weights, and ground-truth definition are unchanged; only the CVE attribute distribution shifts.

The sweep is run on a focused subset of the original capacity-decay study’s grid:  $K \in \{50, 100, 200\}$ ,  $\lambda \in \{1, 3, 12\}$ , 25 evaluation seeds, 6 windows, 5 methods. Total rows: 6,750.

### A. Honest Finding

**The 96.0% per-pair persistence is a synthetic-distribution artifact.** Under real CVE attribute distributions, the overall HygienePrio-full > EPSS-only persistence across all (cell, seed, window) triples falls from 96.0% to **10.0%**. The two methods tie (within 0.5 pp) in 69% of (cell, seed, window) triples; EPSS-only is slightly ahead at  $K = 50$  (cell-mean W6 gaps of 1.3-2.3 pp in its favour) and the two are essentially identical at  $K \geq 100$ . Notably, all 135 HygienePrio wins in the real sweep occur at Window 1 only, per-cell persistence is exactly 0.100 in all 9 cells. This is not a pipeline artifact: after W1 the two methods' score distributions collapse onto the same residual backlog and produce tied selections, so no post-W1 wins are possible.

**Why.** Two reinforcing mechanisms:

(i) *Real EPSS distribution is thinner.* As documented in The HygienePrio scorer, §XII [2] (real  $\Rightarrow$  3.64% EPSS > 0.10 vs. synthetic 12%), real CVE corpora produce a thinner upper-EPSS tail. This makes the EPSS-only baseline less informative in absolute terms but *more concentrated* on the actual ground-truth positives, so EPSS-only's relative discriminative power improves. Under the EEHDA synthetic fixture, EPSS-only ranks 12% of the backlog above the labelling threshold, of which most are noise; under real distributions only 3.64% clear the threshold and they are higher-quality.

(ii) *Real KEV prevalence is much lower.* Real KEV coverage is 0.52% vs. synthetic 8%. The HygienePrio scorer's KEV-recency term contributes far less under real distributions, weakening the quantitative gap between HygienePrio and EPSS-only.

### B. What Survives

**The qualitative capacity-collapse finding is preserved and sharpened.** HygienePrio's W6 P@50 collapses at  $K = 200$  in the real distribution (mean 0.010-0.021 across  $\lambda$ ), *just like EPSS-only's collapse*. The capacity-decay study's H4 rejection (HygienePrio does not retain an absolute floor at high capacity) is reproduced under real distributions and is, if anything, more severe: at  $K = 200$  in the real sweep, all methods drop below 0.025 at W6.

**HRS-only collapses to near-zero under real distributions.** HRS-only's mean W6 P@50 is 0.000-0.002 across all 9 cells. The combination of thinner EPSS tail and lower KEV prevalence means the ground-truth positive set (EPSS > 0.10 AND HRS > 75th percentile) is small to begin with, and HRS-only's rankings cannot find the few positives that exist.

**The compositional decay mechanism is robust.** The capacity-decay study's EPSS decay slope finding (universally negative across the grid) is reproduced. The mechanism, capacity-constrained remediation drains the high-EPSS tail, does not depend on synthetic distribution and is, as expected, more aggressive under thinner real-world tails for the hygiene-augmented method: HygienePrio-full's W6 cell means in the real sweep are uniformly below their synthetic counterparts. EPSS-only is the exception at  $K = 50$ , where its real W6 cell means (0.114-0.121) are actually *higher* than the synthetic ones (0.034-0.053), consistent with the more concentrated real upper-EPSS tail described above.

### C. Operational Implication

Both the HygienePrio scorer's +31 pp synthetic gap (the HygienePrio scorer, §XII [2]) and the capacity-decay study's 96% per-pair persistence are synthetic-distribution effects. The honest cross-paper recommendation is:

- Single-window evaluations like the HygienePrio scorer should report the real-distribution gap (+2.4 pp) as the substantive operational claim; the +31 pp synthetic number is informative about the EEHDA generator, not the field.
- Multi-window evaluations like the capacity-decay study should re-attribute the "HygienePrio wins" claim under real distributions as: the two methods tie (within 0.5 pp) in 69% of (cell, seed, window) triples; EPSS-only is slightly ahead at  $K = 50$  (1.3-2.3 pp at W6) and the two are essentially identical at  $K \geq 100$ , with both methods converging to near-zero at high capacity.
- The structural finding, capacity-driven collapse of any deterministic scoring policy *under self-driven (closed-loop) selection*, is robust across synthetic and real distributions, and is the program's most generalisable result. This matches the theoretical bound proven in a later self-trajectory analysis in this line as the Closed-Loop Signal Exhaustion Theorem. That self-trajectory analysis further shows the collapse is specific to self-driven selection and does not occur under independent drivers.

### D. Honest Caveat on This Section's Scope

The real-distribution test replaces only per-CVE attribute distributions. The fleet topology, host-CVE assignment, HRS components (patch posture, AD exposure, telemetry freshness), ground-truth definition (EPSS > 0.10 AND HRS > 75th percentile), and selection-policy convention are unchanged. A full external validation would require longitudinal real-fleet exploitation outcome data, which remains unavailable in any public dataset. The present section is a partial external-validity bound, not a full replacement for real-data validation.

### E. Reproducibility

The `real_data/` directory lives at the repository root (a sibling of `paper6/`, not inside it):

```
<repo-root>/real_data/real_dist_capacity_sweep.py
<repo-root>/real_data/results/
  real_dist_capacity_sweep.csv
<repo-root>/real_data/results/
  real_dist_capacity_summary.json
```

(6,750-row frozen sweep.) Reproduce from the repository root with `python3 real_data/real_dist_capacity_sweep.py`.

## XI. RELATED WORK

**Single-cell temporal evaluation.** the temporal-stability study [1] established the multi-window evaluation framework at a single cell ( $K = 50, \lambda = 3$ ) and demonstrated EPSS-only decay and HygienePrio retention at that operating point. The present paper generalises that result along the two cell-defining

axes and recovers a qualified picture: relative advantage is robust, absolute floor is not.

**Per-CVE EPSS dynamics.** Ravalico et al. [7] characterised the temporal stability of per-CVE EPSS scores across  $\sim 45,000$  CVEs. Their study examines the random walk of an individual score; the present paper examines the compositional decay of EPSS-as-a-ranking-signal under capacity-constrained remediation. The two effects are independent; the compositional decay documented here would occur even with perfectly stable per-CVE scores.

**Risk-based vulnerability management.** Jacobs et al. [12] demonstrated that exploit-prediction models substantially outperform CVSS-based triage on enterprise data. Their evaluation is at a single window. The present paper does not contest that finding within Window 1 (EPSS-only beats CVSS-only at every cell in our sweep at W1) but adds that the single-window comparative advantage shrinks markedly by W6 and inverts in 6 of 20 cells (all at  $K \in \{25, 50\}$ ).

**Sensitivity studies in prioritization.** Sensitivity analyses of vulnerability-prioritization methods to capacity and arrival rate are uncommon in the open literature; most published comparisons fix both. The present paper documents a case where the choice of  $(K, \lambda)$  would have changed the headline conclusion, and provides a frozen 15,000-row artifact for downstream sensitivity-analysis work.

**Cyber-hygiene benchmarking.** The reproducible hygiene benchmark in this line established the discriminative structure of patch posture and AD group drift in synthetic anomaly-detection tasks. HygienePrio [2] integrated those signals into a scorer. Both are single-window studies. The present paper inherits both as fixed inputs and varies the multi-window operating point.

**Policy mandates.** CISA BOD 22-01 [3] and 23-01 [13], together with NIST SP 800-40 Rev. 4 [4], define the operational window-scheduling regime that the sweep simulates. No prior work, to our knowledge, reports a pre-registered two-dimensional capacity-arrival sweep of prioritization methods under this regime with frozen seeds.

## XII. CONCLUSION

A pre-registered two-dimensional sweep over remediation capacity  $K$  and CVE arrival rate  $\lambda$ , evaluated on 25 seeds across six maintenance windows and five prioritization methods, produces 15,000 result rows that revise the temporal-stability study’s single-cell conclusion in three places.

**First**, EPSS-only Precision@50 decays in every cell of the grid (slope  $-0.025$  to  $-0.052$  P@50/window). No cell studied reaches steady state.

**Second**, the pre-registered monotonicity hypotheses (H1, H2) are rejected: Spearman correlations between cell-mean slope and the swept axes fall well short of the  $|\rho| \geq 0.8$  threshold and frequently carry the wrong sign. The underlying dynamics are non-monotone in the synthetic regime studied, and we report this honestly rather than smoothing.

**Third, and most importantly, the pre-registered claim that HygienePrio-full retains a high absolute floor across**

**all cells (H4) is rejected.** At  $K \in \{100, 200\}$ , HygienePrio-full’s W6 mean P@50 collapses to 0.32 and 0.06-0.12 respectively, not because hygiene signals fail to discriminate but because, under self-driven (closed-loop) selection, high-throughput remediation exhausts the high-HRS tail of the fleet, leaving the residual backlog with little discriminative structure for any host-level signal to recover. A later self-trajectory analysis in this line shows the collapse is specific to self-driven selection and does not occur under independent drivers.

The robustness claim that survives the sweep is per-pair: across all 3,000 (cell, seed, window) comparisons, HygienePrio-full outperforms EPSS-only at P@50 in 96.0% of pairs (minimum per-cell 76.7%). Hygiene augmentation is a strict improvement over EPSS-only across the studied regimes, but is not a guarantee of high absolute scheduling quality at high capacity. Follow-up work in this program, online recalibration (the online-calibration study), multi-history smoothing (the multi-history smoothing study), and self-trajectory evaluation (a later self-trajectory analysis in this line), now reports results across  $(K, \lambda)$  cells rather than at one operating point, and external comparison work should do the same.

**Reproducibility.** The sweep driver, analysis script, pre-registration protocol, and frozen 15,000-row results CSV are available at <https://anonymous.4open.science/r/anonymous>.

**Future work.** The most important next steps are (i) a fully factorial “each method drives its own trajectory” sweep to separate intrinsic EPSS decay from selection-policy coupling, (ii) external validation on real fleet telemetry where exploitation-event logs replace HRS as ground truth, and (iii) integration with non-greedy multi-window scheduling that explicitly models the  $(K, \lambda)$  capacity-arrival balance as a planning input.

## DATA AVAILABILITY STATEMENT

The data and code that support the findings of this study are openly available in the research-papers repository at <https://anonymous.4open.science/r/anonymous>. The manuscript sources, figures, analysis code, and frozen result tables are included and regenerate every reported number deterministically from the fixed seeds.

## DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author.

## FUNDING

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